
CampusMonk
By Rachit Rastogi

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Q - Given a sorted array find a pair whose sum is equal to K.

Example:

INPUT 1
N = 5
Arr[] = {5, 9, 12, 23, 31}
K = 32

OUTPUT1
YES

INPUT 2
N = 6
Arr[] = {1,2,3,4,5,6}
K = 15

OUTPUT 2
NO

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Code

```cpp
#include<bits/stdc++.h>
using namespace std;
int main()
{
    int n;  cin>>n;
    vector<int>arr(n);
    for(int i=0;i<n;i++)
    {
        cin>>arr[i];
    }
    int k;  cin>>k;
    int left = 0, right = n-1;
    while(left<right)
    {
        if(arr[left] + arr[right] == k)
        {
            cout<<left<<" "<<right<<endl;
            return 0;
        }
        else if(arr[left] + arr[right] < k)
        {
            left++;
        }
        else
        {
            right--;
        }
    }
    cout<<"Not found"<<endl;
}
```

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Q - Given a sorted array find the pair whose sum is closest to K(given).

Example:

INPUT1
N = 5
Arr[] = {5, 9, 12, 23, 31}
K = 20

OUTPUT1
9 12

Code

```cpp
#include<bits/stdc++.h>
using namespace std;
int main()
{
    int n;  cin>>n;
    vector<int>arr(n);
    for(int i=0;i<n;i++)
    {
        cin>>arr[i];
    }
    int k;  cin>>k;
    int left = 0, right = n-1;
    int element1 = -1, element2 = -1;
    int possibleAns = INT_MAX;
    while(left<right)
    {
        int sum = arr[left]+arr[right];
        if(abs(sum - k) < possibleAns)
        {
            possibleAns = abs(sum - k);
            element1 = arr[left];
            element2 = arr[right];
        }
        if(sum < k)
        {
            left++;
        }
        else
        {
            right--;
        }
    }
    cout<<element1<<" "<<element2<<endl;
}
```

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(Link in Description)

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Q - Given 2 sorted arrays pick one element from each array and find the sum closest to K(given).

Example:

INPUT1
N = 4 M = 4
Arr1[] = {5, 9, 12, 23}
Arr2[] = {8, 21, 25, 37}
K = 30
OUTPUT1
5 25

INPUT2
N = 6 M = 3
Arr1[] = {2, 4, 6, 10, 19, 24}
Arr2[] = {22, 28, 36}
K = 42
OUTPUT2
6 36

Code

```cpp
#include<bits/stdc++.h>
using namespace std;
int main()
{
    int n,m;  cin>>n>>m;
    vector<int>arr1(n), arr2(m);
    for(int i=0;i<n;i++)
    {
        cin>>arr1[i];
    }
    for(int i=0;i<m;i++)
    {
        cin>>arr2[i];
    }
    int k;  cin>>k;
    int left = 0, right = m-1;
    int element1 = -1, element2 = -1;
    int possibleAns = 999999999;
    while(left < n && right >= 0)
    {
        int sum = arr1[left]+arr2[right];
        if(abs(sum - k) < possibleAns)
        {
            possibleAns = abs(sum - k);
            element1 = arr1[left];
            element2 = arr2[right];
        }
        if(sum < k)
        {
            left++;
        }
        else
        {
            right--;
        }
    }
    cout<<element1<<" "<<element2<<endl;
}
```

Homework
->3 elements ka sum = k

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DOUBTS??